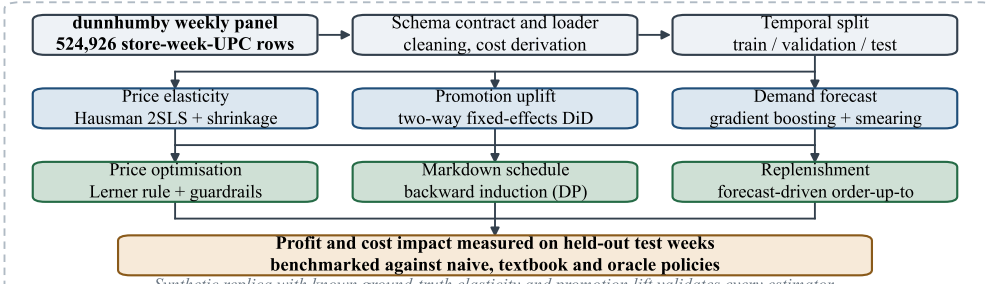
DATA
ESTIMATION
DECISION
EVALUATION
dunnhumby weekly panel
524,926 store-week-UPC rows
Schema contract and loader
cleaning, cost derivation
Temporal split
train / validation / test
Price elasticity
Hausman 2SLS + shrinkage
Promotion uplift
two-way fixed-effects DiD
Demand forecast
gradient boosting + smearing
Price optimisation
Lerner rule + guardrails
Markdown schedule
backward induction (DP)
Replenishment
forecast-driven order-up-to
Profit and cost impact measured on held-out test weeks
benchmarked against naive, textbook and oracle policies
Synthetic replica with known ground-truth elasticity and promotion lift validates every estimator